F2: linear complexity and DFT support of sequences from the p^2-lift of a prime-field curve

Run F2-RUN-02 of study `p2_lift_linear_complexity_20261110`. Class: boundary diagnostic; exact integer computation, no S, no rho row, no leaderboard change. **Status: closed by decision rule 3.** Evidence standing: local evidence; §9a confirmatory EKS run open. All numbers below are `[pcc:unverified]` (no pcc claim id exists for this run). No improvement over rho, or over anything, is claimed.

Question and mechanism

Follow-up to H110 (`p2_lift_defect_20261101`), which closed only the coordinate-section scalar-polynomial bridge. For G of prime order r and $a_k = f([k]G)$, where f is the formal-layer defect of the Hensel section ($\tau_0 = rf$), the sequence is r-periodic over F_p. With $m = \mathrm{ord}_r(p)$, its linear complexity is $L = [\text{mean} \neq 0] + mc$, where c is the number of nonzero degree-m components of the DFT over F_{p^m} and $N = \frac{r-1}{m}$ (Blahut; standard, not formalized). A sparse DFT ($c \ll N$) would make $f([k]G)$ a sparse exponential polynomial in k and transfer the ECDLP to a DLP in $\mu_r \subset F_{p^m}^*$. Question: is $c(f_0)$ or $c(\mathrm{torsion}_u)$ below the null (uniform random periodic sequence) on any curve?

What was run

21 curves: H (the 7 H110 curves), R12, R14, R16 (4 fresh prime-order curves each) and S (2 supersingular curves, $m = 2$); multipliers $\{1, 2, 3, 5\}$; observables O1 f_0, O6 $\mathrm{torsion}_u$ (gated), O2 f_1, O3 $f_1 - f_0$; 200 null sequences per curve; plant O5 on S. 336 rows. Card `CARD-p2-lift-linear-complexity-20261110-F2-RUN-02` (receipt completed, 13 commands, rc 0), pinned head `9465664d`. Local host, 4 CPUs, started 2026-11-11T19:39Z, finished 2026-11-12T01:25Z (20,761.7 s). Tier deviation: the main session ran the card itself (`executor-mechanical` rate-limited). F2-RUN-01 was lost to a container restart (`benchmarks/losses/…f2-run-01…`) and relaunched as RUN-02.

Frozen predictions and rules (PROTOCOL.md §6, §7)

P1 (H, R12, R14, R16): $c(\mathrm{O1}) = c(\mathrm{O6}) = N$, $L \in \{r-1, r\}$. **P2** (S): $c = \frac{r-1}{2}$. **P3** O3 behaves as P1/P2. **P4** O5 gives $L = 2$; O7 drops L by 1 iff mean $\neq 0$. **P5** A/A: identical (L, c) across multipliers. Prior that a gated observable breaks P1 or P2: ≤ 0.02.

1. **Instrument failure:** P4 or P5 fails, or any H110 self-check fails.
2. **Lead alive:** a gated observable has $c < N$ on a curve with $Np^{-m} < 10^{-3}$, or on S $c \leq \frac{r-1}{2} - 3$.
3. **Closed:** otherwise.

Results `[pcc:unverified]`

Source `results/run.json` (21 curves x 4 multipliers x 4 observables). All 336 rows pass every self-check (c integer, two τ_0 splits agree, $\tau(\mathrm{torsion}_u) = 0$, O1 $L_7 = L - [\text{mean} \neq 0]$). **P1, P2, P3 hold:** every observable has $c = N$ exactly and $L \in \{r-1, r\}$ (O1, O2, O3: $L = r - 1$; O6: $L = r$). **P4 holds:** O5 $L = 2$, $c = 1$ on S_0 and S_1. **P5 holds:** identical (L, c) across the four multipliers on 21 of 21 curves. **Empirical null:** 0 of 4,200 random sequences has $c < N$; $c_{\min} = c_{\max} = N$ on every curve. Cells are L/c at multiplier 1; $c/N = 1.0$ for all four observables.

H (7 curves)

curve	p	r	m	N=(r-1)/m	O1 L/c	O6 L/c	O2 L/c	O3 L/c	c/N (all 4)	A/A	null c range (200)	log10 P_vanish
E101	101	83	82	1	82/1	83/1	82/1	82/1	1.0	pass	1..1	−164.35
E127	127	109	108	1	108/1	109/1	108/1	108/1	1.0	pass	1..1	−227.21
E149	149	139	46	3	138/3	139/3	138/3	138/3	1.0	pass	3..3	−99.49
E211	211	223	111	2	222/2	223/2	222/2	222/2	1.0	pass	2..2	−257.69
E283	283	281	70	4	280/4	281/4	280/4	280/4	1.0	pass	4..4	−171.02
E503	503	499	83	6	498/6	499/6	498/6	498/6	1.0	pass	6..6	−223.45
E907	907	947	946	1	946/1	947/1	946/1	946/1	1.0	pass	1..1	−2797.90

R12

curve	p	r	m	N=(r-1)/m	O1 L/c	O6 L/c	O2 L/c	O3 L/c	c/N (all 4)	A/A	null c range (200)	log10 P_vanish
R12_0	4093	3989	1994	2	3988/2	3989/2	3988/2	3988/2	1.0	pass	2..2	−7202.11
R12_1	4093	4153	4152	1	4152/1	4153/1	4152/1	4152/1	1.0	pass	1..1	−14997.20
R12_2	4093	4177	4176	1	4176/1	4177/1	4176/1	4176/1	1.0	pass	1..1	−15083.89
R12_3	4093	4049	2024	2	4048/2	4049/2	4048/2	4048/2	1.0	pass	2..2	−7310.47

R14

curve	p	r	m	N=(r-1)/m	O1 L/c	O6 L/c	O2 L/c	O3 L/c	c/N (all 4)	A/A	null c range (200)	log10 P_vanish
R14_0	16381	16253	16252	1	16252/1	16253/1	16252/1	16252/1	1.0	pass	1..1	−68491.46
R14_1	16381	16453	16452	1	16452/1	16453/1	16452/1	16452/1	1.0	pass	1..1	−69334.33
R14_2	16381	16339	778	21	16338/21	16339/21	16338/21	16338/21	1.0	pass	21..21	−3277.43
R14_3	16381	16363	16362	1	16362/1	16363/1	16362/1	16362/1	1.0	pass	1..1	−68955.04

R16

curve	p	r	m	N=(r-1)/m	O1 L/c	O6 L/c	O2 L/c	O3 L/c	c/N (all 4)	A/A	null c range (200)	log10 P_vanish
R16_0	65521	65843	65842	1	65842/1	65843/1	65842/1	65842/1	1.0	pass	1..1	−317120.13

R16_1	65521	65957	32978	2	65956/2	65957/2	65956/2	65956/2	1.0	pass	2..2	−158834.30
R16_2	65521	65497	8187	8	65496/8	65497/8	65496/8	65496/8	1.0	pass	8..8	−39430.80
R16_3	65521	65701	32850	2	65700/2	65701/2	65700/2	65700/2	1.0	pass	2..2	−158217.80

S ($m = 2$)

curve	p	r	m	N=(r-1)/m	O1 L/c	O6 L/c	O2 L/c	O3 L/c	c/N (all 4)	A/A	null c range (200)	log10 P_vanish
S_0	32693	5449	2	2724	5448/2724	5449/2724	5448/2724	5448/2724	1.0	pass	2724..2724	−5.59
S_1	65171	5431	2	2715	5430/2715	5431/2715	5430/2715	5430/2715	1.0	pass	2715..2715	−6.19

On S, $c = \frac{r-1}{2} = 2724$ and 2715; the rule-2 threshold $\frac{r-1}{2} - 3$ is 2721 and 2712.

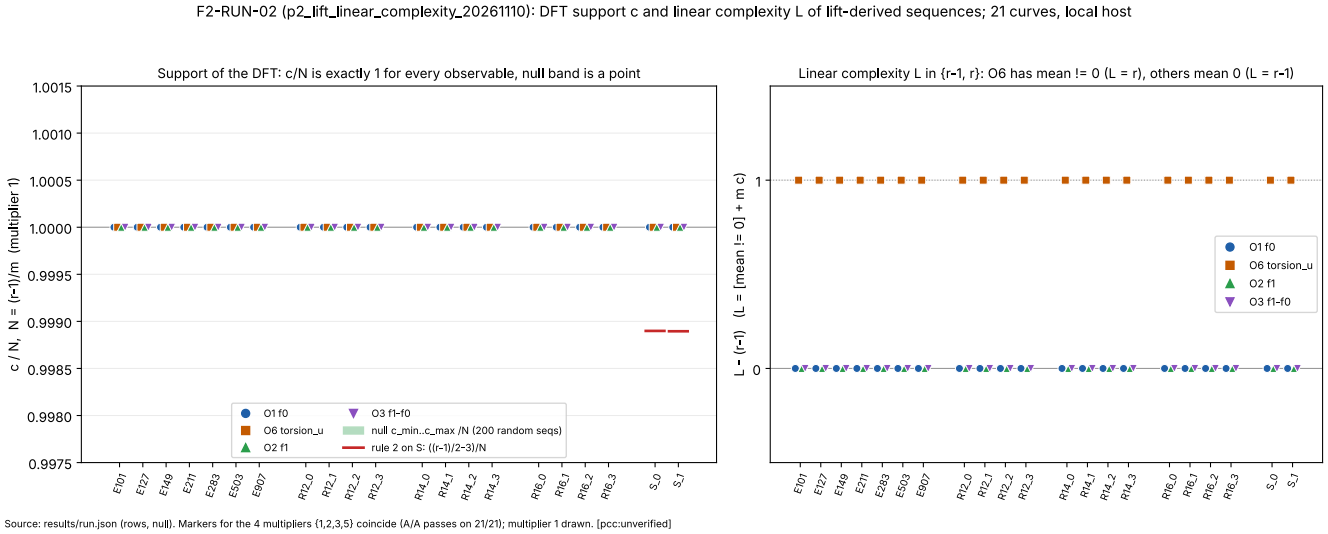


Figure 1: Left: c/N per curve for O1, O6, O2, O3 (all 1.0), the null band $c_{\min}..c_{\max}$ over N (a point at 1) and on S the rule-2 threshold. Right: $L - (r-1)$: 0 for O1, O2, O3 and 1 for O6. Multiplier 1 drawn; the other multipliers coincide. Source: `results/run.json`; `figures/plot.py`.

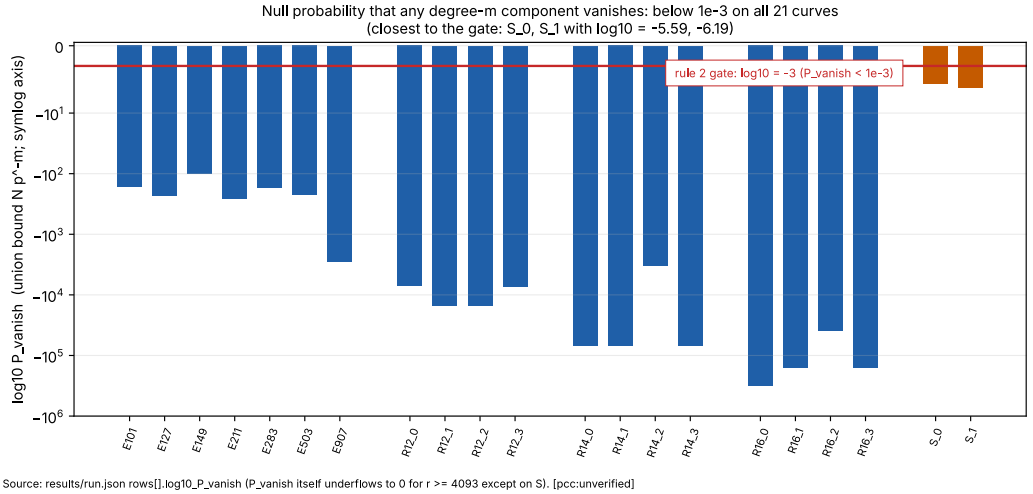


Figure 2: $\log_{10} P_{\text{vanish}}$ (union bound Np^{-m}) per curve against the rule-2 gate 10^{-3} (symlog axis). All 21 curves lie below it; closest: S_0 (−5.59) and S_1 (−6.19).

Cost counts `[pcc:unverified]`

Generation is shared per (curve, multiplier), so it is summed over the 84 pairs, not the 336 rows (row sums are exactly 4.0x): `hensel_lifts` 5,726,624; `inversions_mod_p2` 236,445,680; `inversions_mod_p` 8,589,852; `lifted_adds` 71,525,800; `lifted_doublings` 159,193,256. Berlekamp–Massey over 336 rows: `bm_multiplications` 739,548,609,053; `bm_inversions` 14,315,528 (plant: 43,522 multiplications). Wall time is a shared-host figure, not a result.

Decision (PROTOCOL.md §7, verbatim) `[pcc:unverified]`

Rule 1: no (P4, P5 and all self-checks pass). Rule 2: no (no curve has $c < N$; on S, $c = 2724$ and 2715 are not $\leq \frac{r-1}{2} - 3$). **Rule 3 applies:**

3. **Closed:** otherwise. Reading: the first formal layer's defect has a full DFT support on every curve measured; a sparse-exponential-polynomial transfer through the p^2-lift is closed for the fixed-x section, the $u = 1$ section and the canonical torsion lift's x-digit, at these sizes.

Caveat paragraph (verbatim):

> Rule 3 does not establish a generic lower bound, says nothing about higher formal layers (p^3, H112) or about sections that fix other coordinates, and is a measurement at toy sizes ($r \leq 2^{16}$).

Limitations

For most curves N is 1 to 8, so the test has resolving power mainly on R14_2 ($N = 21$) and S ($N \approx 2700$, where the null vanishing probability is only 2.6×10^{-6} and 6.4×10^{-7}). The four multipliers are an A/A check, not four samples. Local shared host; the isolab-eks confirmatory run (byte diff of `run.json`) is open; tier deviation recorded. Pre-freeze `--quick` runs were disclosed (§11) and not scored. No matched-rho comparison applies; no research-kb query; no novelty claim.

Evidence pointers

Card	sha256 `8dbad5d02c190f8196c17684e8ed59bbd5c8fd009f61c1b863afc1d4ae05bd1a`; receipt `cards/receipts/F2-RUN-02.receipt.json`
Protocol	sha256 `a9e727d29fca88dded3f9f5a2939e32dd23b8e1b10c6b0d93f35a27b6e185ebb`
Tool source	`tool/src/main.rs` sha256 `72dadcb9416a7facb4e6cdb6eb9aa0794ebe646179481d370c4f8d5eaf1d1fe6`
Binary	sha256 `172f1f4dc2db4e0a0d263ad0de4f4439d6d750fa2dce08e0dfe82ebad2ea19d2`
run.json	sha256 `21591ebbafaea0477b4aeb7cd2fdb9caae00c6c7f77902373e90e5332340bf1e` (523,764 bytes)
Tests	7 passed, 0 failed